

Demi Ruohan Wang

demiw@cs.cmu.edu | d3mi.wang | Scholar | LinkedIn | GitHub

Education

Carnegie Mellon University MIIS, School of Computer Science, GPA: 4.08 *Aug 2025 – Dec 2026 (Expected)*

Tongji University B.S. Software Engineering, GPA: 3.95 / 4.0 *Sept 2020 – Jun 2025*

Experiences

Research Intern — Microsoft AI (formerly MSR AI Frontiers Lab) *Jun 2026 – Aug 2026*

- **Web Skill Factory:** Independently built and open-sourced a **self-evolving agent skill pipeline** in [WebWright](#) (6K GitHub stars), distilling task-specific agent scripts into executable Python/Playwright skills at two levels (parameterized workflows and cross-task site primitives), validated through correctness and replay gates.
- Improved held-out WebArena success from **55% to 70% (+15 pp)** and reduced average agent steps from **17.1 to 14.7**; workflow skills ran standalone in **~40s with no LLM calls**, vs. ~10 min from scratch.

Graduate Research Assistant — Carnegie Mellon University (Advisor: Prof. [Graham Neubig](#)) *Sep 2025 – Present*

- **Code-Native Web Agents & Skill Induction [3]:** Demonstrated that hybrid web agents outperform action-only and code-only agents; built a multi-agent **solve-verify-update** pipeline inducing executable Playwright skills, reaching **67.2% on WebArena-Verified (+6.4 pp vs. no-skill baseline)** and revealing that even oracle-correct skills can cause cross-task regressions.

Research Intern — Microsoft Research Asia *Mar 2025 – Jul 2025*

- **RLVR Gradient Dynamics & Adaptive Clipping [1]:** Resolved conflicting findings on token importance through an off-policy-induced gradient-dominance reversal from low- to high-probability tokens; developed **ACPO** in **veRL** with variance-adaptive clipping, ranking first in **15/18 settings** among GRPO-based methods across 3 tasks, 2 regimes, and 3B/7B models.

Research Intern — Ohio State University (Advisor: Prof. [Yu Su](#)) *Apr 2024 – Nov 2024*

- **UGround [2]:** Developed a universal GUI-grounding model (**200K+ downloads**), built a **9M-element** dataset from 773K screenshots, and achieved SOTA results with up to **36% improvement** across web, mobile, and OS benchmarks.

Machine Learning Engineer Intern — ByteDance *Oct 2023 – Feb 2024*

- **Multimodal Content Moderation:** Built annotation and preprocessing pipelines and fine-tuned LLaVA-based VLMs via LoRA on **100K** examples, raising violation-detection precision **62.3% → 90.2%** and reducing manual review by **40%**.

Selected Publications

- [1] Wang P*, **Wang D.***, Luo X., Xu J., Yang X., Feng S., Yang Y., Li D. *What are Key Factors for Updates in RL for LLM Reasoning?* NeurIPS 2026
 - [2] Gou B., **Wang R.**, Zheng B., Xie Y., Chang C., Shu Y., Sun H., Su Y. *Navigating the Digital World as Humans Do: Universal Visual Grounding for GUI Agents.* ICLR 2025 Oral (1.8%)
 - [3] **Wang D.**, Sutawika L., Neubig G. *Skill Induction for Code Agents on Web Automation.* Agent Skills @ CAIS 2026
 - [4] Teng F., Bai J., Yao X., **Wang D.**, Zhao J., Guo Z. *SCRoll: Skip-Connected Rollouts for Mid-Trajectory Rewards in GRPO.* EMNLP 2026 Findings
- * Equal contribution.

Open Source & Projects

Adversarial Verification of LLM-Generated CUDA Kernels *CMU ML Sys Course Project (Solo), 2026*

Built a **four-layer LLM-agent verifier** to generate adversarial tests for benchmark-passing CUDA kernels; identified **8 failure families** and established a **7.5% false-positive lower bound** in Sakana AI's kernel archive, enabling more robust evaluation.

[WebHarbor](#) — Docker-Mirrored Real-World Websites for Web Agents *Open-Source Contributor, 2026*

Audited **5 Docker-mirrored websites** for visual fidelity, functional correctness, and task grounding; built automated **verifiers for 100+ web-agent tasks**, validating solvability, state transitions, and evaluation correctness.

Skills

Languages / Tools	Python, C/C++ , TypeScript, React, Bash, Docker, Git, Linux, Slurm, Azure, AWS
ML / Systems	PyTorch, Transformers, veRL, SkyRL, vLLM, Ray, FSDE, DeepSpeed, Megatron-LM, PEFT/LoRA, W&B
Agents / Evals	Playwright, BrowserGym, OpenHands, Harbor, LiteLLM, MCP, Tool Use, Code/Web/GUI Agents